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# Quantitative Reflection for Mental State Analysis via Aspect Sentiment Triplet Extraction

Kaixi Hu, Lin Li, Kaize Shi, Shaopeng Tang, Haoran Xie, Rui Yan, Guandong Xu

***Abstract*—Aspect Sentiment Triplet Extraction (ASTE) is an important technique for mental state analysis, identifying aspects, opinions, and their sentiment polarities from user-generated content. A fundamental challenge lies in capturing the semantic dependencies across different sentiment elements. Most existing methods are model-centric, primarily modeling triplet-level semantics by aligning predictions with the ground truth. However, users freely post content with varying mental states, showing informal linguistic styles. Such methods may struggle to generalize to challenging cases with element-level diversity. This paper introduces Metamorphic Relation Guided Quantitative Reflection (MaRS), a novel data-centric paradigm that leverages generated testing counterparts of existing samples to improve ASTE. Particularly, inspired by software metamorphic testing, we construct eight metamorphic relations governing intra- and inter-triplet element diversity. Leveraging these relations, we can quantify the weaknesses of existing ASTE methods and large language models (LLMs) in capturing diversity. Furthermore, we devise a quantitative reflection strategy that flexibly incorporates linguistic constraints into both the training and reasoning phases. Extensive experiments on four ASTE datasets and three mental happiness datasets demonstrate that our MaRS outperforms several state-of-the-art models in F1-score and reduces violation rates by up to 10.80% for ASTE models and 17.29% for LLMs.**

***Index Terms*—Aspect sentiment triplet extraction, mental state analysis, element-level diversity, quantitative reflection.**

## I. Introduction

MENTAL health spans a spectrum of psychological states, ranging from nuanced positive and negative sentiments to individual happiness and depression [1–3]. Benefiting from the abundance of user-generated content, a collection of Aspect-Based Sentiment Analysis (ABSA) tasks [4] has rapidly emerged, enabling fine-grained sentiment insights that facilitate individual mental state analysis. The primary objective of ABSA is to extract sentiment elements, including aspect terms, opinion terms, aspect categories, and sentiment polarities. Early studies [5, 6] treated each sentiment element in isolation. Recently, Aspect Sentiment Triplet Extraction (ASTE) [7, 8] has gained considerable attention, posing a great challenge for jointly extracting triplets of aspect terms, opinion terms, and sentiment polarities.

The modeling of semantic correspondence and dependency across sentiment elements is the fundamental challenge. The pioneering work [7] follows a two-stage pipeline. They first introduce two sequence tagging models to predict sentiment elements, and then utilize a classifier to identify aspect-opinion pairs. The subsequent works are developed into unified tagging schemes that extract triplets in one-shot. For example, the sequence tagging [9, 10] captures the dependency between the segments of aspects and opinions. The grid tagging [11–13] jointly encodes element features through table filling. Beyond the tagging schema, some works redefine the original problem as machine reading comprehension [14–17], generative methods [18, 19] and LLM prompting [13, 20]. They all represent promising directions for advancing ASTE.

Despite their success, the above model-centric methods mostly focus on triplet-level semantic modeling by aligning model outputs with the ground truth. Nevertheless, ASTE is a compound sentiment analysis task [4] where element-level understanding are the building blocks of triplet-level semantic modeling. In particular, user-generated content is freely posted with diverse mental states, showing informal linguistic styles. The model-centric paradigm may be insufficient for modeling diverse sentiment elements, thereby limiting existing models' generalization capability. In Fig. 1, we empirically observe the recent deep ASTE model (SLGM [19]) and GPT-4o using Chain-of-Thought (CoT) [13] struggle to handle diverse sentiment expressions in two aspects:

- **Reasoning on Intra-Triplet Elements.** In the first case, after replacing the opinion term "nice" with its synonym "scrummy" [1], SLGM generates a response with the opposite sentiment polarity, whereas GPT-4o-based COT prompting produces the correct response.

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[1]The word "scrummy" is a London speech meaning excellent, marvellous, or delicious. (https://www.oed.com)

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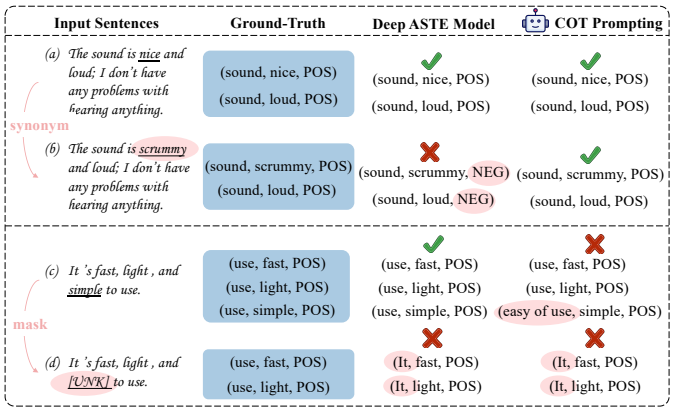



Fig. 1. Responses of the deep ASTE model (SLGM [19]) and GPT-4o [13]. Sentences (b) and (d) are diversified by replacing the underlined opinion terms in sentences (a) and (c) with synonyms or the masking token "[UNK]".

- **Reasoning on Inter-Triplet Elements.** Generally, there are multiple triplets in a sentence. In both cases, replacing an element in one triplet affects SLGM's responses in the other triplets. Moreover, GPT-4o based COT prompting sometimes produces hallucinations [21]. It generates spurious information "easy of", and fails to identify "use" in the other triplets after replacing "simple".

In the above cases, SLGM and COT prompting represent the modeling of expression diversity in the training and reasoning phases, respectively. We observe that LLMs, equipped with rich open-world knowledge, capture diverse elements better. However, it remains limited in addressing the complex diversity across triplets. The limitation of both deep models and LLMs primarily arises from the lack of diverse ASTE samples to guide their modeling process. Building on the data-centric paradigm, some recent work [22, 23] employs data augmentation to increase the diversity of *training samples*. In a complementary direction, our prior work [8] preliminarily generates diverse *testing samples* that conform to the corresponding training batch and leverages their generalization errors to guide the training process. However, exploiting limited ASTE samples to handle complex diversity is still insufficient, especially for cases involving inter-triplet elements. It is necessary to leverage models' reasoning capabilities to capture expression diversity by themselves.

This paper introduces **M**etamorphic Rel**a**tion Guided Quantitative **R**eflection for A**S**TE (MaRS) to support mental state analysis. It provides a comprehensive data-centric solution that employs diverse testing samples to guide both the training and reasoning phases. Particularly, inspired by software metamorphic testing (MT) [24], we construct eight metamorphic relations (MRs) governing intra- and inter-triplet element diversity. These relations quantify the weaknesses in capturing diverse sentiment elements through violation rates (VRs), thereby offering extra signals for modeling diversity.

Furthermore, we propose a quantitative reflection strategy that flexibly injects linguistic constraints in both the training and reasoning settings. For the training-oriented reflection, it leverages VRs to dynamically adjust the loss weights of three element classifiers. For reasoning-oriented reflection, we introduce six reasoning paths based on the identification order of aspects, opinions, and sentiment polarities. Each reasoning path is associated with reflection thoughts that utilize VRs to guide question re-answering. Finally, we conduct metamorphic testing on several ASTE models and LLMs. The results show that existing ASTE models and LLMs still have potential to improve in capturing diverse elements, while our quantitative reflection strategy effectively enhances their capabilities.

To sum up, we make the following contributions:

- This pioneering work highlights the element-level diversity for mental state analysis via ASTE, which serve as the building blocks for triplet-level semantic modeling.
- We propose MaRS, a comprehensive data-centric ASTE solution for mental state analysis. It devises eight metamorphic relations to generate testing samples and quantify weaknesses across intra- and inter-triplet diversity.
- Based on the quantitative results, we propose a quantitative reflection strategy that incorporates linguistic constraints into both the training and reasoning phases.
- Experiments on four benchmark ASTE datasets show that MaRS effectively improves several state-of-the-art ASTE models and LLMs across most reference-based metrics, achieving up to a 17.29% reduction in VRs. Additional evaluation on three mental-happiness datasets further demonstrates its ability to handle diverse expressions.

## II. Related Work

### A. Aspect Sentiment Triplet Extraction

Fine-grained ASTE is crucial for supporting mental state analysis. Most existing methods follow a **tagging schema** first introduced by Peng et al. [7]. They adopted a two-stage pipeline that separately tags sentiment elements. Most subsequent works [9–13, 25, 26] have evolved into end-to-end unified tagging. For instance, sequence tagging works [9, 27–31] enhance the learning of span-based segments. The grid tagging [11] represents all sentiment elements and their relations in a two-dimensional grid. Some studies further improve grid tagging by multi-views [12, 32, 33], boundary tags [25, 26], syntactic information [34] and Transformer [35]. More recently, Sun et al. [13] propose a compact grid tagging scheme, MiniConGTS, by re-evaluating the redundancy.

Other schemas include **machine reading comprehension** (MRC) [14–17], **generative methods** [18, 19, 36] and **LLM prompting methods** [13, 20]. For example, the context-masked strategy [16] and query-oriented adapters [17] are proposed to capture dependencies across multiple aspects and opinions in MRC. The works [19, 36] enhance the generative models by sequence labeling. The above schemas are also transferred across different domains [37–39]. Recently, LLM prompting [13, 20, 40] enhances ASTE with world knowledge.

Despite the effectiveness of above methods, their model-centric paradigm is insufficient to handle diverse sentiment expression due to limited ASTE samples. Some recent work [22, 23] attempt to acquire more training samples through data augmentation, which may suffer from unstable pseudo-label quality. Differently, our previous work [8] preliminarily explores diverse testing counterparts for training batches in

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deep ASTE models. The idea is complementary to traditional data augmentation. However, enhancing the training process alone only improves the model's awareness of diversity within the existing training distribution. And, there are few ASTE training samples available for mental state analysis in practice. The reasoning process plays a more crucial role in generalizing to unseen samples, particularly those exhibiting complex diversity across triplets. This work provides a comprehensive solution that enables model reflection during both the training and reasoning phases.

### B. LLM Prompting-based Reasoning in ABSA

LLMs have demonstrated strong performance across various ABSA tasks [41, 42], attributable to their strong content understanding capabilities. Nevertheless, few studies [13, 40] focus on the ASTE task, as the complex dependencies among sentiment elements across triplets often lead to hallucinations [21, 43]. To address this gap, PASTEL [40] is among the first to attempt finetuning LLMs for ASTE. It separately extracts sentiment elements and refines the final triplets.

Differently, prompting is another direction for optimizing LLMs by carefully designing instructions without modifying its parameters. Some effective prompting strategies include COT [44], self-consistency [45], tree-of-thoughts [46], and graph-of-thoughts [47]. The work [13] takes an early step in applying CoT to ASTE task. HiPM employs a mixture of hard and soft prompts to enhance cross-domain ASTE [38]. However, both finetuned LLMs and CoT remain insufficient for reducing hallucinations. Some recent work [48–50] observes that LLMs can self-correct through reflection questions. Inspired by these studies, our work leverages quantitative results from MRs to guide reflection reasoning on element-level diversity, providing a novel perspective on LLM prompting for ASTE.

### C. Metamorphic Relation in NLP

Metamorphic Testing (MT) [51] is originally designed to address the oracle problem in software engineering, where it is difficult to determine the correctness of outputs. Rather than directly comparing outputs to the ground truth, MT validates a program by examining Metamorphic Relations (MRs) between outputs before and after transformations. This advantage has motivated a line of work to validate NLP models in robustness [52–54], fairness [55, 56], etc.

The basic problem of MT lies in devising MRs tailored to specific tasks and testing purposes. Most existing works [52–54] focus on the machine translation task, emphasizing how morphology and syntax contribute to translation errors. Some works [56–58] study sentiment analysis. However, their sentiment analysis tasks are not fine-grained. Our work differs from existing works in two aspects. First, we devise MRs for fine-grained ASTE task which highlights the linguistic constraints among intra- and inter-triplet elements. Second, whereas prior MT works evaluate NLP models only during the testing phase, our work employs MT as an analysis tool during training or reasoning phases to support quantitative reflection.

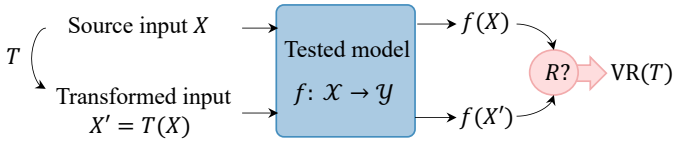



Fig. 2. The workflow of metamorphic testing.

## III. Problem Formulation

### A. Aspect Sentiment Triplet Extraction

This work focuses on fine-grained ASTE for mental state analysis. Given a sentence $X = \{w_1, w_2, \ldots, w_n\}$ with $n$ tokens, the goal is to train a neural network model $f : \mathcal{X} \to \mathcal{Y}$ that extracts $m$ triplets:

$$\mathcal{Y} = \{(a, o, s)_i\}_{i=1}^{m}. \tag{1}$$

In each triplet, $a$ and $o$ are aspect and opinion terms drawn from $X$. The sentiment polarity $s$ of $a$ belongs to $\{\text{POS}, \text{NEU}, \text{NEG}\}$, corresponding to Positive, Neutral, or Negative. $m$ denotes the number of triplets in the sentence. With the identification of fine-grained sentiment elements, we can obtain individuals' mental state analysis, such as happiness prediction, by aggregating these sentiment polarities.

### B. Metamorphic Relation

Investigating the weaknesses of existing models typically requires sufficient sample-pairs that present contrastive conditions while keeping all other factors hygienically controlled. In ASTE, however, annotating all possible aspect–opinion–sentiment triplets is labor-intensive. MT [51] addresses this problem by checking MRs between outputs, which is well-suited for our research objectives.

As shown in Fig. 2, given an input source sentence $X$, a transformation $T : \mathcal{X} \to \mathcal{X}'$ is devised according to established knowledge (e.g., lexical rules), and a transformed sentence $X'$ is derived. After feeding both the sentences into a tested model $f$, MT needs to check whether the outputs satisfy the expected relations:

$$f(T(X)) \sim R(f(X)), \tag{2}$$

where $R : \mathcal{Y} \to \mathcal{Y}'$ defines the corresponding expected transformation of outputs.

Violation Rates (VRs) are a standard metric for quantifying the frequency with which a model fails to satisfy the specified MRs. For a sentence $X_i$, a corresponding transformed case is generated as $X_i' = T(X_i)$. The indicator function $\mathbf{1}(\cdot)$ is used to denote whether a violation occurs. The violation rate of $T$ can be computed, as follows:

$$\text{VR}(T) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\big(f(X_i') \not\sim R(f(X_i))\big) \times 100\%, \tag{3}$$

where $N$ is the number of cases. $f(X_i)$ and $f(X_i')$ are the model outputs for the source and transformed inputs, respectively. Here, $\text{VR}(T) \in [0, 1]$. A higher VR indicates that the model frequently fails to maintain the expected response under input transformations, revealing potential weaknesses.

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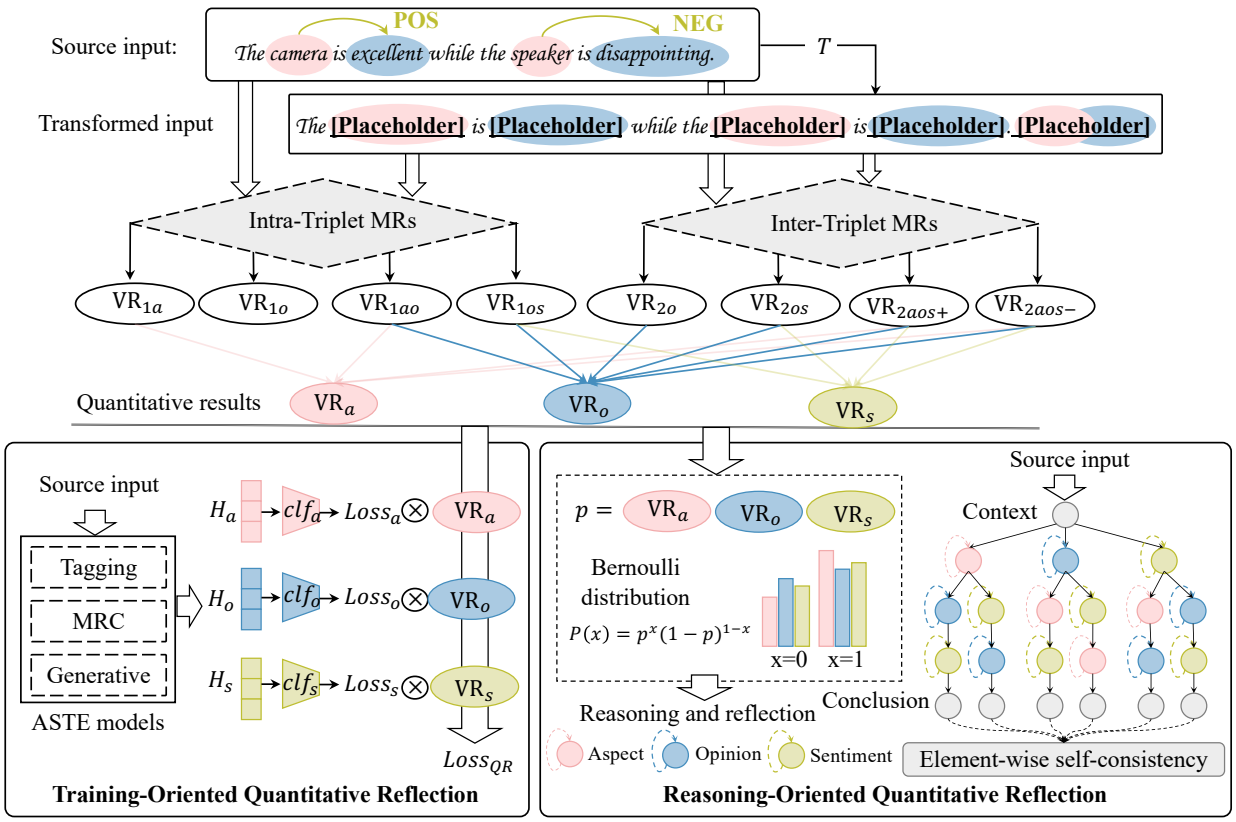



Fig. 3. The proposed MaRS. We follow a data-centric paradigm, where intra- and inter-triplet MRs combined with transformations generate new sentences to quantify weaknesses in capturing element-level diversity. For the training-oriented reflection strategy, it utilize the quantitative results to weight the losses of different sentiment elements. For the reasoning-oriented reflection strategy, the quantitative results are employed to generate a Bernoulli distribution that dynamically adjusts the reflection process across six reasoning paths. The outputs from these paths are then integrated through element-wise self-consistency.

## IV. Methodology

Fig. 3 depicts our proposed MaRS. It first utilizes eight MRs to generate diverse expressions based on the source sentences $X$. With these diverse expressions, it further quantifies the weakness in capturing element-level diversity, and perform the quantitative reflection strategy that flexibly injects linguistic constraints in both training and reasoning phases.

### A. Metamorphic Relation Construction

To quantify the weaknesses in capturing element-level diversity, we devise eight MRs. Specifically, these MRs consider the correspondence and dependency over intra-triplet elements and inter-triplet elements. We illustrate them using "(camera, excellent, POS)" as a pivot in Table I.

*1) MRs for Intra-Triplet Element Diversity:* Intra-triplet element diversity arises from the variations among aspect terms, opinion terms, and sentiment polarities within a triplet. We design MRs that apply transformations to the aspect and/or opinion terms. The corresponding elements in the output triplet are updated accordingly, while the remaining elements are unchanged. We have four intra-triplet MRs, as follows:

**Definition 1** (Aspect Synonym or Hypernym, $\mathrm{MR}_{1a}$)**.** *Given that sentiment is context-dependent, we consider modifying the sentiment holder by applying synonym or hypernym transformations to the aspect term. Consequently, the aspect in the pivot should be updated (e.g., "camera" → "device"), while the remaining elements are unchanged.*

**Definition 2** (Opinion Synonym, $\mathrm{MR}_{1o}$)**.** *Linguistic expressions provide diverse ways to convey similar meanings. To adapt to this variability, we apply synonym transformations to the opinion term in the pivot triplet. The opinion in the output triplet is modified accordingly (e.g., excellent → fantastic), while the remaining elements remain unchanged.*

**Definition 3** (Aspect and Opinion Synonym , $\mathrm{MR}_{1ao}$)**.** *Transforming only the aspect or opinion is insufficient. Hence, we transform both simultaneously using synonyms or hypernyms. For expected responses, the aspect and opinion terms should be updated (e.g., "camera" → "device", "excellent" → "fantastic"), while the sentiment polarity remains unchanged.*

**Definition 4** (Opinion Antonym, $\mathrm{MR}_{1os}$)**.** *Antonym transformations change the meaning of the source input, requiring the model to understand the revised semantics and accurately determine the corresponding sentiment polarity. In the expected responses, the opinion is updated accordingly, the aspect remains unchanged, and the sentiment polarity is reversed.*

*2) MRs for Inter-Triplet Element Diversity:* Typically, a sentence contains multiple triplets. Triplets beyond the pivot exhibit diversity in aspects, opinions, and sentiment polarities, which can influence the prediction of the pivot triplet. A model

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TABLE I
THE EXAMPLES OF MRS AND THEIR TRANSFORMATIONS.

| Transformation Types | | Contents | Expected Responses |
|---|---|---|---|
| Source sentence | | The camera is excellent while the speaker is disappointing. | (camera, excellent, POS)<br>(speaker, disappointing, NEG) |
| Intra-triplet | $\text{MR}_{1a}$ | The device is excellent while the speaker is disappointing. | camera → device |
| | $\text{MR}_{1o}$ | The camera is fantastic while the speaker is disappointing. | excellent → fantastic |
| | $\text{MR}_{1ao}$ | The device is fantastic while the speaker is disappointing. | camera → device, excellent → fantastic |
| | $\text{MR}_{1os}$ | The camera is terrible while the speaker is disappointing. | excellent → terrible, POS → NEG |
| Source sentence | | The camera is excellent while the speaker is disappointing. | (camera, excellent, POS)<br>(speaker, disappointing, NEG) |
| Inter-triplet | $\text{MR}_{2o}$ | The camera is excellent while the speaker is poor. | (camera, excellent, POS)<br>keeps unchanged |
| | $\text{MR}_{2os}$ | The camera is excellent while the speaker is fantastic. | |
| | $\text{MR}_{2aos+}$ | The camera is excellent while the speaker is disappointing. Overall, the phone is unsatisfactory. | |
| | $\text{MR}_{2aos-}$ | The camera is excellent while the [UNK] is [UNK]. | |

[1] Without loss of generality, we illustrate the intra-triplet and inter-triplet MRs using the triplet "(camera, excellent, POS)" as a pivot.
[2] We marked the transformations of aspect terms in red and opinion terms in blue.
[3] For simplicity, the column "expected responses" shows variations of the pivot "(camera, excellent, POS)", with other elements unchanged.

with adequate capacity to handle such variability is expected to mitigate these effects. Consequently, inter-triplet MRs are designed to capture the influence between the pivot triplet and the surrounding triplets.

**Definition 5** (Opinion Synonym, $\text{MR}_{2o}$)**.** *Each triplet's prediction is expected to remain independent. Synonym transformations applied to the opinions of other triplets should not affect the pivot triplet. For example, in Table I, replacing "disappointing" with a synonym leaves the pivot triplet's prediction unchanged.*

**Definition 6** (Opinion Antonym, $\text{MR}_{2os}$)**.** *Similar to synonym transformations, inverting the sentiment polarity of other triplets through antonym transformations should not affect the prediction of the pivot triplet.*

**Definition 7** (Add Aspects and Opinions, $\text{MR}_{2aos+}$)**.** *We further introduce diversity by adding additional interference. Specifically, we append a new sentence to the source sentence. This appended sentence contains a triplet with sentiment opposite to that of the pivot triplet. These additions are not expected to affect the extraction of the pivot triplet.*

**Definition 8** (Mask Aspects and Opinions, $\text{MR}_{2aos-}$)**.** *In addition to introducing additional interference, we mask the semantic information of other triplets by replacing their aspect and opinion terms with "[UNK]". The pivot triplet's output is expected to remain consistent with its original prediction, despite the removal of this contextual information.*

*3) Considerations Behind MRs:* The above MRs are constructed by combining different sentiment elements (i.e., aspect, opinion, and sentiment polarity). Other combinations can also be considered, such as replacing all intra-triplet elements (e.g., $\text{MR}_{1aos}$) or replacing aspects across inter-triplets (e.g., $\text{MR}_{2a}$, $\text{MR}_{2ao}$, $\text{MR}_{2aos}$). However, these combinations present worse sensitivity to uncovering models' weakness in our Wilcoxon signed-rank test (see Section V-E1). The possible reasons are as follows:

- Directly replacing all elements (e.g., $\text{MR}_{1aos}$) can easily lead to uncertain semantic variations, leading to a inaccurate VR. Hence, when considering multiple triplets (e.g., $\text{MR}_{2aos+}$), a paraphraser (see Section V-C) is utilized to append new sentence.
- For multiple triplets, the aspects typically focus on distinct entity attributes or dimensions. Therefore, replacing aspects across inter-triplets may offer limited benefit.

In addition, some language toolkits or dictionaries (see implementation details in Section V-E1) can be utilized, providing rules to guarantee the validity of the transformations. During the transformation process, we also generate multiple candidate options and select those aspect–opinion combinations that frequently co-occur for replacement.

*4) VRs Computation:* With the definitions of MRs, we can calculate the corresponding violation rates $\text{VR}_i$ using the Eq. (3), where $i \in \{1a, 1o, 1ao, 1os, 2o, 2os, 2aos+, 2aos-\}$. These VRs reflect the weakness of the prediction models in coping with diverse sentiment expressions, and will be leveraged to devise our quantitative reflection strategy.

### B. Quantitative Reflection Strategy

The ASTE task can be approached in both the training and reasoning phases. However, focusing on only one of these phases is insufficient for capturing complex diversity, particularly in cases involving inter-triplet elements. As such, we propose a quantitative reflection strategy to flexibly inject linguistic constrains into these phases by leveraging the VRs.

*1) Weakness Quantitation:* The VRs reflect the weakness in capturing diversity of different elements. According the

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elements affecting by MRs, we aggregate the VRs into aspect-related $\text{VR}_a$, opinion-related $\text{VR}_o$, and sentiment polarity-related $\text{VR}_s$. Their definitions are below:

$$\text{VR}_e = \frac{1}{|g_e|} \sum_{i \in g_e} \text{VR}_i, e \in \{a, o, s\}, \tag{4}$$

where the groups are $g_a = \{1a, 1ao, 2aos+, 2aos-\}$, $g_o = \{1o, 1ao, 1os, 2o, 2os, 2aos+, 2aos-\}$, and $g_s = \{1os, 2os, 2aos+, 2aos-\}$. These aggregated VRs reveal quantitative weaknesses in capturing aspect terms, opinion terms, and sentiment polarities. They are subsequently leveraged to inform our quantitative reflection strategy for training and reasoning, respectively.

*2) Training-Oriented Quantitative Reflection:* We first consider to use the quantitative results of weakness to enhance the training phases of existing models. Specifically, a diversity-aware loss is proposed to weight the identification of elements. And then, this loss is further integrated with the principal ASTE loss for holistic training.

*a) **Diversity-Aware Loss**:* Given an input sentence $X$ containing $m$ triplets, and since the details of model structures are agnostic, we extract the representations of aspects, opinions, and sentiment polarities from the final hidden layer. Their representations can be denoted below:

$$H_e = \{h_{e_1}, h_{e_2}, \ldots, h_{e_m}\}, e \in \{a, o, s\}. \tag{5}$$

To encourage training models to better engage in learning these representations, we simply introduce three classifiers (e.g., three-layer MLPs) with sigmoid activation functions, to identify different sentiment elements, as follows:

$$\hat{y}_e = \text{Sigmoid}(\text{clf}_e(H_e)), e \in \{a, o, s\}. \tag{6}$$

And then, three binary cross-entropy (BCE) losses can be derived, as follows:

$$\mathcal{L}_e = \text{BCE}(y_e, \hat{y}_e), e \in \{a, o, s\}, \tag{7}$$

where $y_e$ is the ground-truth, with all its values set to 1. Finally, we use the quantitative results, i.e., $\text{VR}_a$, $\text{VR}_o$, $\text{VR}_s$, as the weights of the BCE losses, and obtain the diversity-aware loss as follows:

$$\mathcal{L}_{div} = \sum_{e \in \{a,o,s\}} \text{VR}_e \mathcal{L}_e. \tag{8}$$

*b) **Holistic Training Loss**:* Let $\mathcal{L}_{aste}$ denote the principal loss of a training model, typically computed as the cross-entropy between the ground-truth and predicted results. Inspired by the work [59], they automatically weight multi-task losses by considering how uniform the discrete distributions are. We approximate the holistic loss as follows:

$$\mathcal{L}_{hol} \approx \frac{1}{\sigma_1^2} \mathcal{L}_{aste} + \frac{1}{\sigma_2^2} \mathcal{L}_{div} + \log \sigma_1 + \log \sigma_2, \tag{9}$$

where $\sigma_1$ and $\sigma_2$ are automatically learned during the training phase. In practice, we adopt the improved version [60], designed to prevent negative loss values.

*3) Reasoning-Oriented Quantitative Reflection:* The annotation of multiple sentiment elements is a labor-intensive process, resulting in the scarcity of ASTE training data. Consequently, some studies [13, 20] attempt to approach the ASTE task during the reasoning phase. However, these methods remain insufficient for mitigating hallucinations. Our quantitative reflection strategy enhances the reasoning process by reflection-enhanced multiple path reasoning and element-wise self-consistency.

*a) **Reflection-Enhanced Multiple Path Reasoning**:* To encourage reasoning step by step, we decompose ASTE into three subtasks, i.e., aspect reasoning, opinion reasoning, and sentiment polarity reasoning. Since the order of element extraction may influence the final results, we follow the idea of Self-Consistency [45], and reorganize the subtasks into six reasoning paths, as illustrated in the lower-right corner of Fig. 3. These reasoning paths share the same thoughts of context and conclusion. Table II presents the instruction contents. Specifically, the sentence $X$ is embedded into the template at the beginning "$\$text$" during the reasoning of aspects, opinions, and sentiment polarities. And, the element types and their corresponding contents extracted from the previous steps are both embedded into the template at "$\$pre_element_type$" and "$\$pre_element$", which establish a connection between different reasoning steps.

Notably, the reflection thought facilitates the self-correction of errors in the reasoning process. However, the reasoning quality varies across different sentiment elements. Excessive reflection may reduce reasoning efficiency, whereas insufficient reflection can lead to poor performance. Hence, we incorporate the quantitative results (i.e., $\text{VR}_a$, $\text{VR}_o$, and $\text{VR}_s$) into the probability $p$ of Bernoulli distributions as follows:

$$P(x) = p^x (1-p)^{1-x}, p \in \{\text{VR}_a, \text{VR}_o, \text{VR}_s\}, \tag{10}$$

where $P(x)$ denotes the probability mass function of the binary variable $x$, with $x \in \{0, 1\}$. When $x = 1$, the model engages in reflection on the reasoning of the element, whereas $x = 0$ indicates that reflection is not applied. Therefore, under this setting, the larger the quantitative results, the higher the likelihood that reflection is performed.

*b) **Element-Wise Self-Consistency Aggregation**:* We construct six reasoning paths, each yielding a distinct result, which are subsequently aggregated to obtain the final prediction. Unlike Self-Consistency [45], where aggregation relies on majority voting, ASTE requires handling nested elements within triplets, for which simple voting may fail to ensure consistency. As such, we propose element-wise self-consistency based on the longest common subsequence of words (LCSW).

Our element-wise self-consistency is outlined in Algorithm 1. It compares the triplets from different reasoning paths (lines 2–5) and applies LCSW to extract the longest common words of each element (lines 6–7). If the sentiment polarities of the longest common opinions match, the corresponding triplet is included in the final results (lines 8–11).

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TABLE II
INSTRUCTION TEMPLES FOR LLMS.

| Thought Type | Instruction Contents |
|---|---|
| Context | You are an expert in information extraction and sentiment analysis. Please analyze the given text for aspect sentiment triplet extraction. Aspect is an attribute or entity. It is usually a noun or a noun phrase derived from text. Opinion is a descriptive term or phrase that expresses sentiment polarity towards a certain aspect. It is usually an adjective or a descriptive phrase derived from text. Sentiment is a polarity of opinion expressed towards a certain aspect, only as positive, neutral or negative. Let's think step by step. |
| Aspect/Opinion reasoning | The text is: $\$text$. You have previously identified the $\$pre_element_type$ of the text: $\$pre_element$. Please identify the $\$element_type$ in the given text, which may contain more than one. Note that $\$element_type$ is. derived from text. Please output in this format: $\langle$ "$\$element_type$" $\rangle$ |
| Sentiment polarity reasoning | The text is: $\$text$. You have previously identified the $\$element_type$ of the text: $\$element$. Please identify the sentiment polarity that may be present in the given text, which may contain more than one. Note that sentiment is. one of positive, neutral and negative. Please output in this format: $\langle$ "$\$element_type$" $\rangle$ |
| Reflection | Please think again about $\$element_type$ in text, especially focusing on span margins for aspect and opinion or the consistency between sentiment and the pair. Note that aspect and opinion are derived from text, and sentiment is one. of positive, neutral and negative. Please output in this format: $\langle$ "$\$element_type$" $\rangle$ |
| Conclusion | According to the thought process, please conclude the triplet in text and return it in the format [("aspect", "opinion", "sentiment")]. Each triplet must contain an aspect, an opinion, and the corresponding sentiment polarity. There may be more than one triplet in a text, please return them completely, returning them in the format: [("aspect1", "opinion1", "sentiment1"), ("aspect2", "opinion2", "sentiment2")]. |

**Algorithm 1:** Element-wise Self-Consistency

**Input:** Predicted Triplets $\{\mathcal{Y}_1, \ldots, \mathcal{Y}_K\}$ from $K$ reasoning paths, where $\mathcal{Y}_k = \{(a, o, s)_i\}_{i=1}^{m_k}$
**Output:** Final aggregated triplets: $\mathcal{Y}_o$

Initialize a duplicate-free set $\mathcal{Y}_o \leftarrow \emptyset$;
**for** $\mathcal{Y}_{k1} \in \{\mathcal{Y}_1, \ldots, \mathcal{Y}_K\}$ **do**
  **for** $\mathcal{Y}_{k2} \in \{\mathcal{Y}_{k1+1}, \ldots, \mathcal{Y}_K\}$ **do**
    **for** $t1 \in \mathcal{Y}_{k1}$ **do**
      **for** $t2 \in \mathcal{Y}_{k2}$ **do**
        $com_a \leftarrow$ LCSW($t1[0]$, $t2[0]$);
        $com_o \leftarrow$ LCSW($t1[1]$, $t2[1]$);
        **if** $(com_a \neq \emptyset)$ *and* $(com_o \neq \emptyset)$ *and* $(t1[-1] = t2[-1])$ **then**
          temp $\leftarrow (com_a, com_o, t1[-1])$;
          **if** $temp \notin \mathcal{Y}_o$ **then**
            $\mathcal{Y}_o \leftarrow \mathcal{Y}_o \cup temp$
**return** $\mathcal{Y}_o$;

## V. EXPERIMENTS

To investigate the efficacy of our proposed MaRS, extensive experiments are conducted to answer the following questions:

- **RQ1:** Are reference-based evaluations (e.g., F1-score) sufficient to reflect a model's capability in capturing expression diversity?
- **RQ2:** Can VRs be used to compare the capability in capturing expression diversity across different models?
- **RQ3:** How do the variants of our quantitative reflection strategy perform under different combinations of key components?

TABLE III
DATASET STATISTICS.

| | Type | LAP14 | RES14 | RES15 | RES16 |
|---|---|---|---|---|---|
| Train | # Sentences | 906 | 1,266 | 605 | 857 |
| | # POS Triplets | 817 | 1,692 | 783 | 1,015 |
| | # NEU Triplets | 126 | 166 | 25 | 50 |
| | # NEG Triplets | 517 | 480 | 205 | 329 |
| Dev. | # Sentences | 219 | 310 | 148 | 210 |
| | # POS Triplets | 169 | 404 | 185 | 252 |
| | # NEU Triplets | 36 | 54 | 11 | 11 |
| | # NEG Triplets | 141 | 119 | 53 | 76 |
| Test | # Sentences | 328 | 492 | 322 | 326 |
| | # POS Triplets | 364 | 773 | 317 | 407 |
| | # NEU Triplets | 63 | 66 | 25 | 29 |
| | # NEG Triplets | 116 | 155 | 143 | 78 |

### A. Dataset Description

The early ASTE datasets are derived from the SemEval Challenges [61, 62], covering the laptop (LAP) and restaurant (RES) domains. This work adopts the widely used ASTE-Data-V2 [63], which includes annotations where a single opinion span is associated with multiple targets. The dataset statistics are summarized in Table III.

Notably, the number of sentences is relatively small, and triplets with negative sentiment polarity are fewer than those with positive polarity. Consequently, the diversity of these datasets is limited, which aligns with our motivation to enhance the capability in capturing element-level diversity.

### B. Baselines and Evaluation Metrics

Our MaRS is model-agnostic. It can be integrated into most existing models. As discussed in the taxonomy of existing methods in Section II, we apply MaRS to fourteen state-of-the-art methods, and conduct comparisons against them. The details are as follows:

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- **Tagging methods**: EMCGCN [12] and BDTF [25] adopt grid tagging, whereas SimSTAR [9] and STAGE [10] rely on sequence tagging. MiniConGTS [13] represents the advanced tagging methods.
- **MRC methods**: MRC methods have gradually been replaced by generative and LLM-based reasoning. COM-MRC [16] is one of the widely adopted baselines.
- **Generative methods**: It represents a promising direction for advancing ASTE. GAC [64] and MCL [65] are the state-of-the-art methods. Since their source code is unavailable, we directly report the results from their papers without integrating our MaRS. SLGM [19] serves as the most widely adopted baseline.
- **LLM prompting methods**: LLM prompting baselines conduct ASTE in the reasoning phase, including Prompt [20], CoT [13], and PASTEL [40]. Note that, in addition to prompting, PASTEL employs supervised fine-tuning (SFT) to further enhance LLM performance. Following its SFT strategy, we optimize Llama3.2-1B and perform ASTE reasoning using our prompts.
- **Data-centric methods**: T2S [22] and GeDa [23] are both data augmentation methods for ASTE. T2S relies on a sentence generator, whereas GeDa leverages LLMs for augmentation. For a fair comparison, we reproduce GeDa on the ASTE-Data-V2 dataset rather than the version refined by the authors.

Following prior works [13, 16], we report model performance by comparing its predictions with the ground truth using reference-based metrics, i.e., precision (P), recall (R), and F1-score. Unless stated otherwise, the results of the baselines are from their original papers. We particularly discuss the limitations of reference-based metrics to evaluate the reasoning capabilities regarding diversity in Section V-D. Hence, we further explore the utilization of VRs to perform model testing.

### C. Implementation Details

For each synonym, antonym, or hypernym transformation, up to 10 candidates are extracted from NLTK [2] or an online dictionary [3]. And, Parrot [4] is employed to automatically generate a new sentence appended to the source sentence for $\text{MR}_{2aos+}$. For training-oriented reflection, we apply MaRS to the baseline models using the hyper-parameters reported in their original works. The reasoning-oriented reflection is conducted on Llama3.1-8B and QWen2-7B without SFT, and on Llama3.2-1B with SFT. We define six reasoning paths and use the violation rates (i.e., VR$a$, VR$o$, and VR$_s$) as the probability $p$ in the Bernoulli distributions. After applying MaRS [5], each method is executed five times with different random seeds, and the mean performance is reported. All deep models are trained on an NVIDIA TITAN XP GPU.

### D. Reference-based Performance Comparison (RQ1)

Extensive experiments are conducted to compare MaRS with a range of baselines using reference-based metrics.

[2]https://www.nltk.org
[3]https://github.com/meetDeveloper/freeDictionaryAPI
[4]https://github.com/PrithivirajDamodaran/Parrot_Paraphraser
[5]The code will be released later.

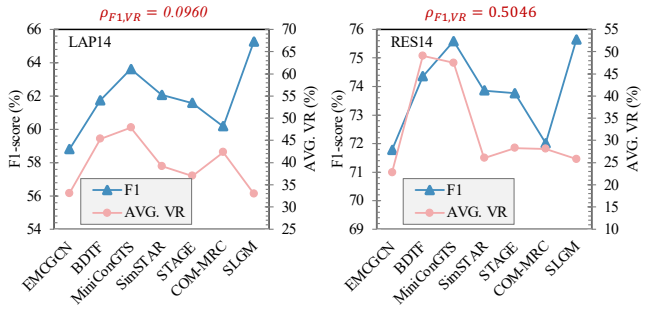



Fig. 4. The Pearson correlation between F1-scores and the average of eight VRs of ASTE models without MaRS.

*1) Comparison with Deep Model-Based ASTE:* The upper part of Table IV reports the comparison results with deep model–based ASTE methods. In addition, we plot the variations in F1 and the average of the eight VRs, and compute their Pearson correlation coefficient in Fig. 4. The training time is shown in Table V. We have the following findings:

**MaRS enhances existing ddep models across a majority of reference-based metrics.** In Table IV, integrating MaRS into various ASTE models yields higher F1 (red parts), with few exceptions (blue parts). Notably, the F1 improvements primarily stem from higher recall and the declined difference between precisions and recalls. This observation can be attributed to MaRS's capability to improve the perception of element-level diversity. However, while modeling diversity increases the number of candidate spans for MRC-based models, the reduction in precision outweighs the gain in recall, thereby resulting in a small decline of F1.

**Reference-based metrics is limited to reflect model performance.** First, in Table III, due to the labor-intensive nature of triplet annotation, both the training and testing sets contain fewer than a thousand sentences. With such limited test samples, the results of reference-based evaluation seem not much convinced. Second, we typically expect that a strong model achieves high F1 alongside low VRs. However, the observation from Fig. 4 is opposite to our expectation. And, the Pearson correlation coefficients show a weak positive correlation, indicating the insufficiency of reference-based metrics to reflect a model's capability.

**MaRS introduces a moderate amount of additional computational overhead.** To assess the impact of MaRS on training time, we compare the training time of the best baseline model SLGM before and after the integration of MaRS. As shown in Table V, we find that MaRS leads to an increase in training time ranging from 4.81% (on the REST16 dataset, from 1040s to 1090s) to 6.80% (on the REST14 dataset, from 1604s to 1713s).

*2) Comparison with LLM Prompting-Based ASTE:* Prompting uses carefully designed instructions to improve LLMs' reasoning capabilities. The lower part of Table IV reports the comparison results with prompting methods. We observe that both simple I/O prompting and CoT applied to LLMs without SFT yield inferior performance compared with our MaRS. This is because LLMs inherently possess extensive knowledge of diversity, and our reasoning-oriented quantitative reflection

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TABLE IV
OVERALL PERFORMANCE COMPARISON MODEL-CENTRIC METHODS (%).

| | Method | LAP14 | | | | RES14 | | | | RES15 | | | | RES16 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | P | R | F1 | $\Delta F1$ | P | R | F1 | $\Delta F1$ | P | R | F1 | $\Delta F1$ | P | R | F1 | $\Delta F1$ |
| Deep model-based ASTE | **Tagging methods** | | | | | | | | | | | | | | | | |
| | EMCGCN [12] | **61.70** | 56.26 | 58.81 | +0.89 | 71.21 | 72.39 | 71.78 | +0.78 | 61.54 | **62.47** | 61.93 | +0.15 | 65.62 | **71.30** | 68.33 | +0.09 |
| | + MaRS | 60.26 | **59.15** | **59.70** | | **71.70** | **73.45** | **72.56** | | **61.89** | 62.27 | **62.08** | | **65.88** | 71.15 | **68.42** | |
| | BDTF [25] | **68.94** | 55.97 | 61.74 | +0.96 | **75.53** | 73.24 | 74.35 | +0.19 | 68.76 | 63.71 | 66.12 | +1.33 | 71.44 | 73.13 | 72.27 | +0.65 |
| | + MaRS | 66.46 | **59.33** | **62.70** | | 75.15 | **73.94** | **74.54** | | **69.67** | **65.36** | **67.45** | | **71.75** | **74.12** | **72.92** | |
| | MiniConGTS [13] | **66.82** | 60.68 | 63.61 | +1.10 | **76.10** | 75.08 | 75.59 | +0.64 | **66.50** | 63.86 | 65.15 | +0.70 | **75.52** | 74.14 | 74.83 | +0.38 |
| | + MaRS | 66.00 | **63.46** | **64.71** | | 74.38 | **78.20** | **76.23** | | 65.37 | **66.34** | **65.85** | | 74.83 | **75.61** | **75.21** | |
| | SimSTAR [9] | **66.46** | 58.23 | 62.07 | +0.68 | **76.23** | 71.63 | 73.86 | +0.30 | **71.71** | 59.59 | 65.09 | +0.18 | **72.02** | 74.12 | 73.06 | +0.64 |
| | + MaRS | 65.64 | **60.08** | **62.75** | | 75.20 | **73.14** | **74.16** | | 70.14 | **61.03** | **65.27** | | 71.55 | **75.98** | **73.70** | |
| | STAGE [10] | **71.98** | 53.86 | 61.58 | +1.31 | 78.58 | 69.58 | 73.76 | +0.93 | **73.63** | 57.90 | 64.79 | +0.22 | **76.67** | 70.12 | 73.24 | +0.65 |
| | + MaRS | 68.99 | **57.79** | **62.89** | | **78.95** | **70.91** | **74.69** | | 71.35 | **59.73** | **65.01** | | 75.70 | **72.16** | **73.89** | |
| | **MRC methods** | | | | | | | | | | | | | | | | |
| | COM-MRC [16] | **62.35** | 58.16 | **60.17** | -0.27 | **75.46** | 68.91 | **72.01** | -0.34 | **68.35** | 61.24 | **64.53** | -0.41 | **71.55** | 71.59 | **71.57** | -0.59 |
| | + MaRS | 59.58 | **60.23** | 59.90 | | 73.65 | **69.80** | 71.67 | | 65.99 | **62.38** | 64.12 | | 68.90 | **73.21** | 70.98 | |
| | **Generative methods** | | | | | | | | | | | | | | | | |
| | GAC [64] | 69.14 | 59.82 | 64.14 | +1.28 | 79.25 | 73.82 | 76.44 | -0.10 | 70.40 | 67.54 | 68.94 | +0.03 | 76.95 | 75.80 | 76.37 | +0.37 |
| | MCL [65] | 66.47 | 60.96 | 63.59 | +1.83 | 73.87 | 72.72 | 73.29 | +3.05 | 65.55 | 68.80 | 67.14 | +1.83 | 72.71 | 76.37 | 74.79 | +1.95 |
| | SLGM [19] | **70.54** | 60.74 | 65.27 | +0.15 | 78.84 | 72.70 | 75.64 | +0.70 | 69.75 | **66.85** | 68.27 | +0.70 | 75.86 | 75.76 | 75.80 | +0.94 |
| | + MaRS | 70.11 | **61.33** | **65.42** | | **79.01** | **73.84** | **76.34** | | **71.21** | 66.80 | **68.97** | | **76.45** | **77.04** | **76.74** | |
| LLM prompting-based ASTE | **Llama3.1-8B w/o SFT** | | | | | | | | | | | | | | | | |
| | + I/O Prompting [20] | 26.65 | 27.44 | 27.04 | | 25.82 | 39.64 | 31.27 | | 8.31 | 35.67 | 13.48 | | 8.14 | 42.02 | 13.64 | |
| | + COT [13] | 28.90 | 29.65 | 29.27 | +6.71 | 42.60 | 37.63 | 39.96 | +9.30 | 38.43 | 42.47 | 40.35 | +5.23 | 42.37 | 45.91 | 44.07 | +7.13 |
| | + MaRS | **37.15** | **34.89** | **35.98** | | **52.32** | **46.53** | **49.26** | | **44.63** | **46.56** | **45.58** | | **52.05** | **50.36** | **51.20** | |
| | **QWen2-7B w/o SFT** | | | | | | | | | | | | | | | | |
| | + I/O Prompting [20] | 23.45 | **26.52** | 24.89 | | 22.57 | **37.93** | 28.30 | | 7.52 | 35.46 | 12.41 | | 7.63 | 43.00 | 12.96 | |
| | + COT [13] | 18.72 | 15.10 | 16.72 | +4.45 | 16.15 | 19.01 | 17.47 | +14.41 | 7.44 | 24.74 | 11.44 | +31.96 | 6.16 | 24.71 | 9.86 | +35.66 |
| | + MaRS | **33.92** | 25.86 | **29.34** | | **51.74** | 36.37 | **42.71** | | **48.06** | **41.20** | **44.37** | | **52.92** | **44.96** | **48.62** | |
| | **Llama3.2-1B w/ SFT** | | | | | | | | | | | | | | | | |
| | + PASTEL [40] | 67.83 | 62.61 | 65.22 | +0.80 | 81.95 | 79.81 | 80.87 | +0.31 | 72.54 | 71.18 | 71.86 | +0.43 | 80.89 | 77.43 | 79.12 | +0.56 |
| | + MaRS | **69.59** | **62.80** | **66.02** | | **82.32** | **80.08** | **81.18** | | **73.05** | **71.55** | **72.29** | | **81.63** | **77.82** | **79.68** | |

TABLE V
TRAINING TIME (S) BEFORE AND AFTER INTEGRATING MARS

| | LAP14 | REST14 | REST5 | REST16 |
|---|---|---|---|---|
| SLGM | 1604 | 1132 | 761 | 1040 |
| SLGM+MaRS | 1713 | 1188 | 804 | 1090 |

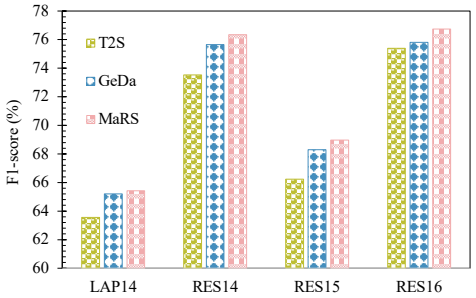



Fig. 5. Comparison of data-centric methods incorporated into SLGM [19].

enables them to identify their weaknesses and more effectively leverage this knowledge. Moreover, following the fine-tuning strategy of PASTEL [40], our MaRS consistently achieves the best performance, indicating the superiority of our reflection.

*3) Comparison with Data-Centric Methods:* Since SLGM [19] is the best deep model-based method with publicly available source code, we incorporate different data-centric methods (e.g., T2S [22] and GeDa [23]) into it for comparison. We observe that both GeDa and our MaRS improve the performance of SLGM, whereas T2S leads to a performance decline. The underlying reason is that synthetic training data generated by T2S and GeDa may lead to inconsistent pseudo-label. Although GeDa filters part of the inconsistent data, our MaRS still achieves superior performance. This is because we employ generated testing data to enhance the learning of existing training data, which avoid the direct impact of inconsistent data.

Overall, MaRS improves existing models in terms of recall and F1. However, reference-based evaluations are insufficient to fully capture a model's capability to handle diversity. Models with high F1 often produce incorrect responses when subjected to synonym, antonym, or hypernym transformations. It is necessary to explore effective metrics that reflect a model's capability in capturing diversity.

### *E. MRs-based Capability Comparison for Diversity (RQ2)*

*1) Effectiveness of MRs vs. Reference-based Testing:* To investigate whether our MRs are sensitive to the model's capability to handle diversity, we introduce the following mutants in line with the work [66]:

- **Gaussian Fuzzing (GF)**: It adds Gaussian noise to the weights of selected neurons.
- **Weight Shuffling (WS)**: It shuffles the incoming weights of randomly chosen neurons to disrupt their connection patterns with the previous layer.
- **Neuron Effect Blocking (NEB)**: It removes the selected neurons by resetting their outgoing weights to zero.

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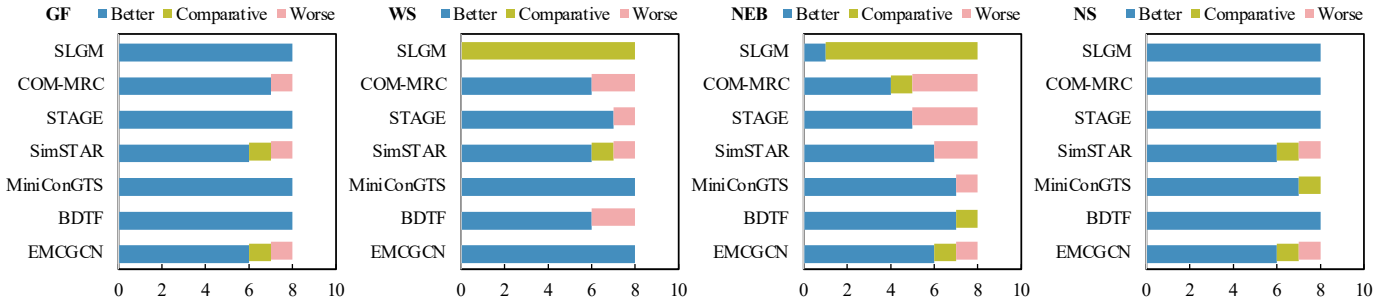



Fig. 6. Sensitivity comparison between eight MRs and F1-scores based on the Wilcoxon signed-rank test. Blue bars represent cases where MRs are more sensitive than F1-scores, and red bars represent the reverse. LLMs are excluded for better comparison, since they performs well in terms of F1-scores.

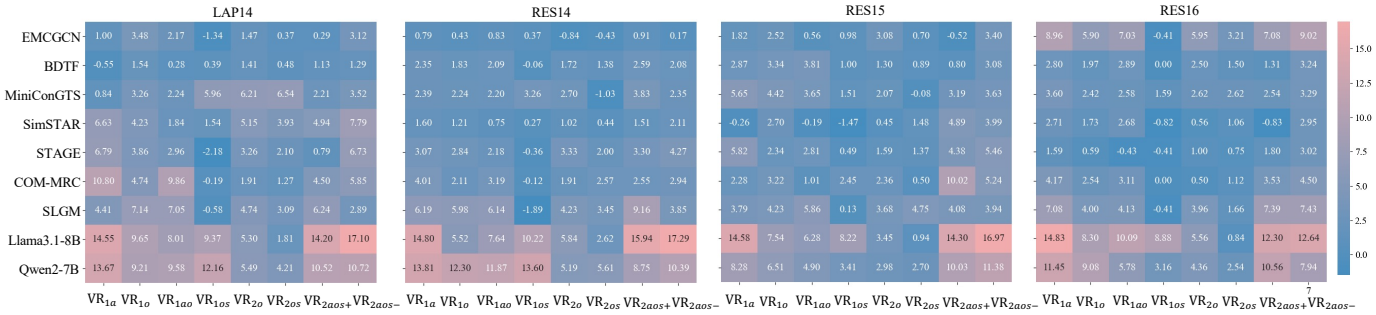



Fig. 7. Heatmaps of the decline of VRs (%) after integrating with our MaRS on four datasets. Higher values, approaching red, indicate a greater decline and thus a better capability for handling diversity.

- **Neuron Switch (NS)**: It switches the weights of two neurons in the same layer.

Following the work [67], we can calculate a pairs of VRs for each mutant where the source sentences are the passed cases of the original ASTE model under reference-based testing and MR-based MT, respectively. Then, the Wilcoxon signed-rank test are used to compare their sensitivity for diversity.

Specifically, the Wilcoxon signed-rank test includes both two-tailed and one-tailed analyses to compute the $p$-values at the significance level $\delta = 0.05$. For the two-tailed test, if $p \geq \sigma$, the sensitivity of MRs and reference-based testing is comparable. For the one-tailed test, if $p < \sigma$, the sensitivity of MRs is better; otherwise, it is worse. The statistics of the Wilcoxon signed-rank test are presented in Fig. 6. Across most cases, blue bars dominate, indicating that the proposed MRs demonstrate greater sensitivity than F1-scores in detecting mutant weaknesses. The few "worse" outcomes occur primarily under WS and NEB, likely due to the inherent sensitivity of the seven Transformer-based deep models.

*2) Capability Gain for Diversity:* Our discussion in the previous subsection V-E1 validates that VRs are effective in evaluating the capability to handle expression diversity. Therefore, to investigate how our MaRS boosts the modeling of diversity, we use heatmaps in Fig. 7 to illustrate the decline of VRs after introducing MaRS. For reasoning-oriented quantitative reflection, we compare the VRs between COT [13] and our MaRS. The prompting is based on LLMs without SFT, focusing on the effect of reasoning process.

We have the following three findings: *First*, LLM prompting exhibits a substantial decline (up to 17.29% in RES14) while ASTE models decline by up to 10.80% in VRs across most situations, which aligns with the conclusions drawn from the F1-values. *Second*, for COM-MRC, unlike the decline in F1-values shown in Table IV, the VRs decrease, indicating improved capability in handling diversity. *Third*, some VRs exhibit a slight increase after introducing our MaRS (as reflected by negative values in the heatmap), particularly for $\text{VR}_{1os}$. This is because most deep model–based ASTE methods rely on complex label modeling, which is insufficient for capturing antonymic opinion transformations. In contrast, MiniConGTS employs a contrastive strategy that explicitly learns opinion reversals, thereby maintaining a decline in $\text{VR}_{1os}$. *Finally*, by averaging the VR reductions, we observe that $\text{MR}_{2aos-}$, $\text{MR}_{1a}$, and $\text{MR}_{2aos+}$ contribute the most, with mean reductions of 5.93%, 5.75%, and 5.23%, respectively. In contrast, the contributions of $\text{MR}_{2os}$ and $\text{MR}_{1os}$ are smaller, with mean reductions of 1.91% and 2.13%. Despite their varying magnitudes, all of them contribute to the overall VR reduction of capturing sentiment elements.

### F. Ablation Study (RQ3)

Based on the quantitative results from MRs, we devise a quantitative reflection strategy for training and reasoning. In particular, the training-oriented reflection consists of a single component (i.e., diversity-aware loss), whose variant corresponds to remove the entire MaRS in Table IV. In contrast, the reasoning-oriented reflection integrates two key prompting components, i.e., reflection-enhanced multi-path reasoning and element-wise self-consistency. To investigate their effectiveness, we set the following variants:

- **Non-Reflective**: It removes the Bernoulli distribution-based quantitative reflection.

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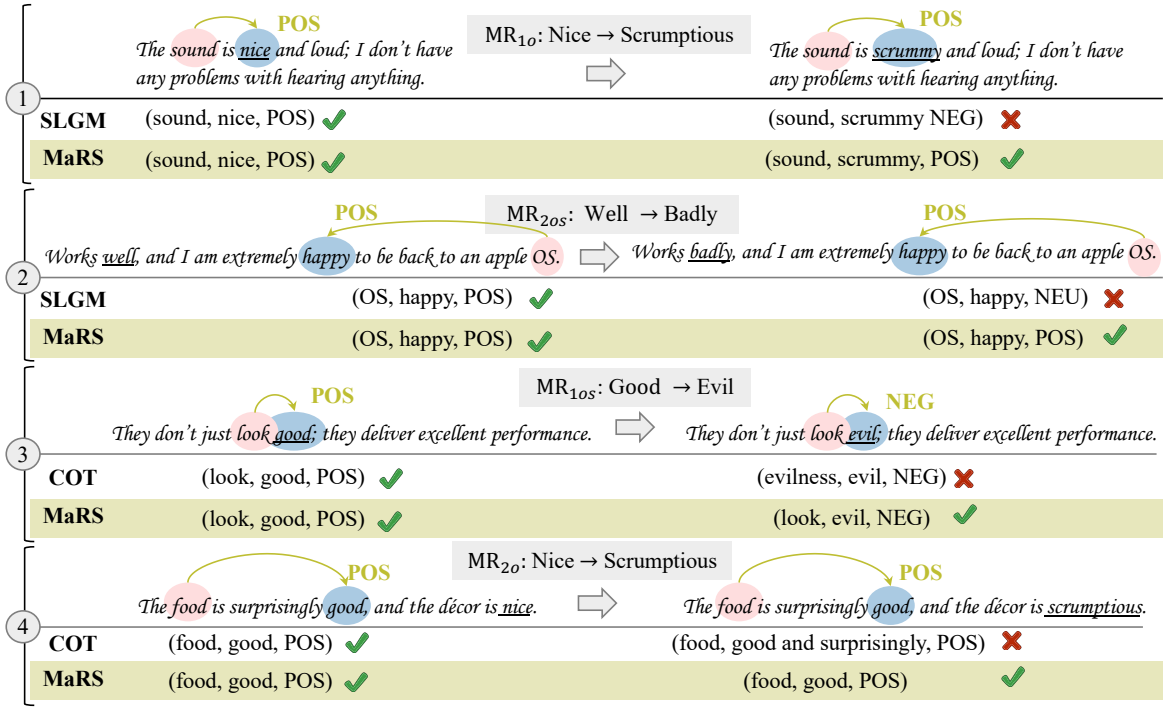



Fig. 8. Case study on diverse sentiment expressions. Cases ① and ② compare our MaRS with SLGM [19] under intra- and inter-triplet element diversity, respectively. Similarly, cases ③ and ④ compare our MaRS with CoT [13] in promoting LlaMA-3.1-8B.

TABLE VI
THE ABLATION RESULTS OF OUR REFLECTION STRATEGY

| LLMs | Variants | LAP14 | | | REST14 | | |
|---|---|---|---|---|---|---|---|
| | | P | R | F1 | P | R | F1 |
| Llama3.1-8B | MaRS | **37.15** | **34.89** | **35.98** | **52.32** | **46.53** | **49.26** |
| | Non-Reflective | 35.94 | 29.31 | 32.29 | 48.65 | 44.78 | 46.63 |
| | Single_Path | 15.79 | 25.04 | 19.36 | 23.71 | 30.97 | 26.86 |
| | Voting | 30.42 | 31.47 | 30.94 | 43.25 | 38.03 | 40.47 |
| Qwen2-7B | MaRS | **33.92** | **25.86** | **29.34** | **51.74** | **36.37** | **42.71** |
| | Non-Reflective | 32.67 | 21.53 | 25.63 | 49.23 | 33.42 | 39.81 |
| | Single_Path | 11.95 | 19.54 | 14.82 | 21.84 | 27.42 | 24.31 |
| | Voting | 20.45 | 22.38 | 21.37 | 33.90 | 30.28 | 31.99 |

- **Single_Path**: We have six reasoning paths, i.e., "a-o-s", "a-s-o", "o-a-s", "o-s-a", "s-a-o", "s-o-a". Since their performances are close, we report the average of the reference-based metrics..
- **Voting**: It integrates the responses from different paths through voting [45], rather than using our element-wise self-consistency. We report the best results achieved with a minimum voting threshold of 2.

Table VI presents the results of variants. Our findings are as follows: *First*, our proposed MaRS consistently outperforms all variants, demonstrating the effectiveness of both reflection-enhanced multi-path reasoning and element-wise self-consistency. *Second*, the variant with a single path present the worst performance, highlighting the contribution of multiple reasoning paths. *Third*, compared with REST14, the sentiment polarities in the LAP14 test set are more balanced. The variant with voting enables LLMs to achieve higher recall on LAP14, indicating the coordinated contributions of reflection and multi-path reasoning.

### G. Case Study

To intuitively explain the efficacy of our MaRs in handling challenging cases with element-level diversity, we transform and analyze several sentences based on MRs in Fig. 8.

Our findings are as follows: *First*, our proposed MaRS, the recent SLGM [19] and COT-based LLM are all show correct responses in the source sentences in Cases ①-④. *Second*, Case ① replaces the opinion term "nice" with its synonym "scrumptious", while Case ③ replaces "good" with its antonym "evil". For these transformed sentences, MaRS still produces correct responses, whereas SLGM and CoT fail to identify the sentiment polarity and aspect term, respectively. This demonstrates that MaRS effectively enhances models in handling intra-triplet element diversity. *Third*, a similar pattern is observed in Cases ② and ④. We modify other triplets within the sentences, yet only MaRS produces the correct response for the pivot triplet. This highlights MaRS's capability to enhance models in handling inter-triplet element diversity.

### H. Application to Mental State Analysis

Mental state analysis is a typical application of ASTE. We further apply our MaRS to extract triplets from social survey data, and then utilize these triplets to predict individuals' happiness state [68, 69]. Specifically, European Social Survey (ESS) [6] and Student Survey (SS) [7] and Chinese Social Survey (CSS) [8] describe different aspects of an individual, including income, lifestyle, and family. Each aspect involves responses

[6]https://ess-search.nsd.no/
[7]https://www.kaggle.com/datasets/hopesb/student-depression- dataset
[8]http://cgss.ruc.edu.cn/

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TABLE VII
THE PREDICTION RESULTS OF MENTAL HAPPINESS STATE (%).

| LLMs | Variants | ESS | | SS | | CSS | |
|---|---|---|---|---|---|---|---|
| | | P | F1 | P | F1 | P | F1 |
| Llama3.1-8B | MaRS | **68.00** | **68.51** | **83.00** | **82.00** | **70.00** | **70.52** |
| | Non-Reflective | 62.00 | 59.96 | 81.00 | 80.00 | 66.00 | 68.01 |
| | Single_Path | 49.00 | 51.37 | 67.00 | 65.57 | 52.00 | 55.16 |
| | Voting | 60.00 | 58.35 | 65.00 | 61.76 | 64.00 | 69.86 |
| | I/O Prompt | 36.00 | 32.14 | 54.00 | 49.03 | 22.00 | 22.09 |
| Qwen2-7B | MaRS | **72.00** | **72.12** | **82.00** | **81.06** | **47.00** | **54.26** |
| | Non-Reflective | 63.00 | 60.81 | 80.00 | 79.68 | 43.00 | 43.10 |
| | Single_Path | 54.00 | 54.42 | 67.00 | 65.57 | 35.00 | 43.09 |
| | Voting | 57.00 | 57.48 | 79.00 | 78.65 | 41.00 | 43.72 |
| | I/O Prompt | 20.00 | 20.98 | 58.00 | 56.89 | 19.00 | 21.97 |

from individual participants. However, these datasets annotate individuals only with an overall mental happiness score (from unhappy to happy) rather than providing sentiment polarity for each aspect. Therefore, we preliminarily aggregate the extracted sentiment polarities of different aspects through voting to infer whether an individual is happy.

Table VII presents the results of mental happiness state prediction. We have the following findings: *First*, MaRS still consistently outperforms all variants, showing the effectiveness of reasoning-oriented quantitative reflection for capturing diversity. *Second*, both variants "Non-Reflective" and "Single_Path" show performance degradation, indicating the enhancement of our reflection for multi-path reasoning. *Third*, the efficacy on both benchmark datasets and mental happiness datasets demonstrates that our MaRS can generalize to different scenarios, as it enhances the generalization capability of existing models by emphasizing element-level diversity.

The success of mental happiness prediction demonstrates the potential of ASTE in supporting mental state analysis. Nevertheless, predicting mental happiness captures only one aspect of the broader spectrum of mental states. Our proposed MaRS framework can be extended to investigate a variety of psychological conditions and traits, including depression, stress, anxiety, loneliness and emotional regulation, highlighting its versatility for comprehensive mental state assessment.

## VI. CONCLUSION AND FUTURE WORK

This paper proposes a novel data-centric paradigm, metamorphic relation guided quantitative reflection for mental state analysis via ASTE, termed MaRS. It first employs eight MRs to quantify the weakness of models for handling diverse sentiment expressions. Then, based on the quantitative results, we further propose a reflection strategy to enhance the training and reasoning phases. The experiments demonstrate that models enhanced with our MaRS not only achieve superior performance on reference-based metrics (e.g., F1-scores) but also reduce the violation rates (VRs) across different MRs, indicating their capability in handling diversity.

In future work, we will integrate the training- and reasoning-oriented quantitative reflections to jointly enhance existing models. And, our MaRS is currently limited to leveraging linguistic constraints for devising MRs, leaving opportunities for extending the framework to multimodal mental state analysis. These MRs can also be used to achieve data augmentation, facilitating model training.

## VII. ETHICAL CONSIDERATIONS

The datasets used in our experiments mainly involve user-generated content from online platforms. They are widely accepted within the academic community, and the private information has already been removed. Consequently, they inherently present a low risk of containing offensive content. Moreover, the analysis results may provide auxiliary support for sellers or clinicians in enhancing their services. However, they should not be regarded as definitive determinations or substitutes for professional judgment.

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